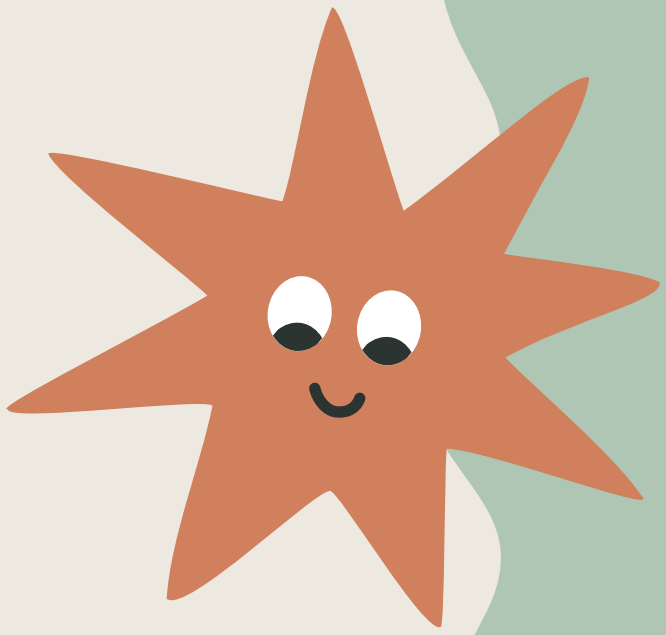
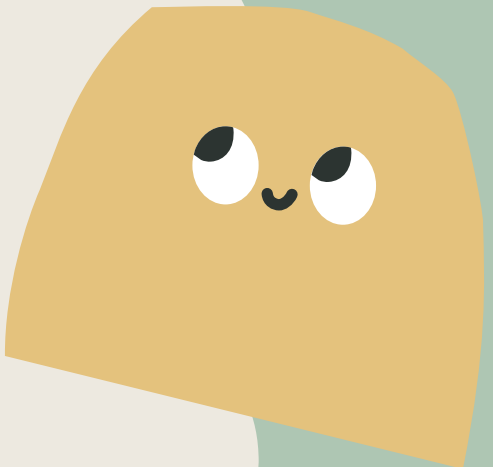
Introduction to Python

Lecture 1





Learning Objectives

- By the end of this lecture, students should be able to:
  - Explain what Python is
  - Understand why Python is widely used in AI
  - Identify major applications of Python
  - Understand Python's basic characteristics
  - Understand Python interpreter and source code at a basic level
  - Install Python and VS Code
  - Write and execute their first Python program

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What is Python?

- Python is a high-level, general-purpose programming language known for its simple syntax and wide range of applications
- Python was created by **Guido van Rossum** and its first public release was in **1991**
- Python allows us to give instructions to a computer using relatively simple and readable syntax
- For example:

```
print("Hello World")
```

- Python allows us to focus more on **the problem we want to solve** and less on complicated syntax

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Why Are We Learning Python?

- We are not learning Python just because Python is popular.
- We are learning Python because it is one of the most important tools used to implement Artificial Intelligence and Machine Learning
- Python is widely used in:
  - Artificial Intelligence
  - Machine Learning, Deep Learning
  - Data Science, Automation
  - Web Development, Computer Vision
  - Natural Language Processing
  - Generative AI

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Why Python is Popular in AI

1. Simple syntax
2. Huge library ecosystem
   - Instead of implementing everything from scratch, programmers can use existing libraries
   - NumPy → numerical computing
   - Pandas → data processing
   - Matplotlib → visualization
   - Scikit-learn → machine learning
   - PyTorch → deep learning
   - OpenCV → computer vision

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Why Python is Popular in AI

3. Excellent for data processing
   - AI requires a lot of data
   - Python provides powerful tools for:
   - Data → Processing → Analysis → Visualization → AI Model
4. Strong AI/ML community
   - There are huge number of tutorials, libraries, GitHub projects, courses, documentation for Python
5. Used in real-world AI
   - Python is widely used in AI research and development
   - So learning Python gives you a practical foundation for the rest of your AI course

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Important Characteristics of Python

- High-level
  - don't have to manage computer hardware details directly
- Interpreted
  - Python programs are executed by the Python interpreter
- Dynamically typed
  - We don't normally have to declare the variable's data type explicitly
- Object-oriented
  - Python supports classes and objects
- Cross-platform
  - Python programs can generally run on Windows, Linux, macOS
- Open Source
  - Python is freely available and has a large community

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Python Versions and Installation

- Python 2 is an older version.
- Modern Python development uses Python 3
- I recommend to install “Anaconda”
- Check Python installation `python --version`
- Python Bash (the Python interactive interpreter)
- Demonstrate Anaconda, VS Code, Jupyter Notebook
- Print() statement and comments

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